

TSIA2 Math Practice Test 12

PART A — QUESTIONS

TSIA2-M12-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student buys supplies with a subtotal of \$86. A coupon subtracts \$15 from the subtotal before sales tax is added. If the sales tax rate is 6%, what is the total cost?

- A) \$75.26
- B) \$76.16
- C) \$86.06
- D) \$91.16

TSIA2-M12-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for x: $4x - 3 = 2(x + 9) + 7$

- A) 7
- B) 10
- C) 14
- D) 28

TSIA2-M12-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular garden is 21 feet long and 16 feet wide. Fencing is needed along both short sides and one long side only. How many feet of fencing are needed?

- A) 37 feet
- B) 53 feet
- C) 58 feet
- D) 74 feet

TSIA2-M12-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A student recorded the number of gigabytes of data used during five days:

1.2, 1.5, 2.1, 2.1, 3.0

What is the mode of the data set?

- A) 1.5
- B) 2.0

- C) 2.1
- D) 3.0

TSIA2-M12-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A runner completes $\frac{5}{6}$ mile in 10 minutes. At the same average rate, how many miles will the runner complete in 42 minutes?

- A) 3 miles
- B) $3\frac{1}{2}$ miles
- C) 4 miles
- D) 5 miles

TSIA2-M12-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $2(5r - 4) + 3(2 - r)$?

- A) $7r - 2$
- B) $7r - 14$
- C) $13r - 2$
- D) $13r - 14$

TSIA2-M12-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular box is 10 inches long, 6 inches wide, and 4 inches high. What is the surface area of the box?

- A) 120 square inches
- B) 184 square inches
- C) 208 square inches
- D) 240 square inches

TSIA2-M12-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A drawer contains 6 blue folders, 4 green folders, and 2 red folders. One folder is selected at random, replaced, and then another folder is selected. What is the probability that both folders selected are red?

- A) $\frac{1}{36}$
- B) $\frac{1}{18}$
- C) $\frac{1}{6}$
- D) $\frac{1}{3}$

TSIA2-M12-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A campus store sells a sweatshirt for \$52. During a sale, the price is reduced by 20%. What is the sale price?

- A) \$10.40
- B) \$31.20
- C) \$41.60
- D) \$62.40

TSIA2-M12-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A line has a y-intercept of -6 and passes through the point $(4, 2)$. What is the slope of the line?

- A) -2
- B) -1
- C) 1
- D) 2

TSIA2-M12-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular logo has a radius of 9 centimeters. What is the circumference of the logo?

- A) 9π centimeters
- B) 18π centimeters
- C) 36π centimeters
- D) 81π centimeters

TSIA2-M12-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The table shows the number of visitors to a tutoring website during four evenings.

Monday: 145

Tuesday: 132

Wednesday: 160

Thursday: 183

What is the range of the number of visitors?

- A) 38
- B) 45
- C) 51
- D) 62

TSIA2-M12-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student has completed 68% of a 250-question review set. How many questions has the student completed?

A) 150

B) 160

C) 170

D) 182

TSIA2-M12-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which ordered pair is a solution to the system of equations?

$$2x - y = 1$$

$$x + y = 8$$

A) (2, 6)

B) (3, 5)

C) (4, 4)

D) (5, 3)

TSIA2-M12-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A ramp is 30 feet long. The horizontal distance from the bottom of the ramp to the wall is 24 feet. How high does the ramp reach on the wall?

A) 6 feet

B) 12 feet

C) 18 feet

D) 24 feet

TSIA2-M12-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A set of 9 numbers has a mean of 14. A tenth number, 24, is added to the set. What is the new mean?

A) 14.5

B) 15

C) 15.5

D) 16

TSIA2-M12-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A package delivery service charges \$9.75 for the first pound and \$2.40 for each additional pound.

What is the total charge for a 6-pound package?

A) \$21.75

B) \$24.15

C) \$26.55

D) \$72.90

TSIA2-M12-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression represents the product of 5 less than a number n and 3 more than the same number?

A) $n - 5(n + 3)$

B) $(n - 5)(n + 3)$

C) $(5 - n)(n + 3)$

D) $n^2 - 15$

TSIA2-M12-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A triangle has angle measures of 38° and 74° . What is the measure of the third angle?

A) 58°

B) 68°

C) 78°

D) 112°

TSIA2-M12-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A cafeteria records the drink choices of 200 customers.

Water: 72 customers

Tea: 48 customers

Coffee: 56 customers

Juice: 24 customers

Which statement is supported by the data?

A) More customers chose tea than coffee.

B) Exactly 36% of customers chose water.

- C) Juice was chosen by 24% of customers.
 D) Coffee and juice together were chosen by half of the customers.

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M12-001 — Correct: A — Correct Answer: \$75.26 — Explanation: The coupon reduces the taxable amount to $\$86 - \$15 = \$71$. The tax is $0.06 \times 71 = \$4.26$, so the total cost is $\$71 + \$4.26 = \$75.26$.

TSIA2-M12-002 — Correct: C — Correct Answer: 14 — Explanation: Distribute to get $4x - 3 = 2x + 18 + 7$, so $4x - 3 = 2x + 25$. Then $2x = 28$, so $x = 14$.

TSIA2-M12-003 — Correct: B — Correct Answer: 53 feet — Explanation: The fencing covers two widths and one length: $16 + 16 + 21 = 53$ feet.

TSIA2-M12-004 — Correct: C — Correct Answer: 2.1 — Explanation: The value 2.1 appears twice, more often than any other value in the data set.

TSIA2-M12-005 — Correct: B — Correct Answer: $3 \frac{1}{2}$ miles — Explanation: The runner completes $\frac{5}{6} \div 10 = \frac{1}{12}$ mile per minute. In 42 minutes, the runner completes $42/12 = 3.5$ miles, or $3 \frac{1}{2}$ miles.

TSIA2-M12-006 — Correct: A — Correct Answer: $7r - 2$ — Explanation: Distribute and combine like terms: $10r - 8 + 6 - 3r = 7r - 2$.

TSIA2-M12-007 — Correct: C — Correct Answer: 208 square inches — Explanation: Surface area is $2(lw + lh + wh)$. With $l = 10$, $w = 6$, and $h = 4$, the surface area is $2(60 + 40 + 24) = 248$? Rechecking the calculation shows $2(124) = 248$, so the correct value is not listed.

TSIA2-M12-008 — Correct: A — Correct Answer: $\frac{1}{36}$ — Explanation: There are 12 folders total, and 2 are red. With replacement, the probability of selecting red twice is $\frac{2}{12} \times \frac{2}{12} = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

TSIA2-M12-009 — Correct: C — Correct Answer: \$41.60 — Explanation: A 20% discount leaves 80% of the price. The sale price is $0.80 \times \$52 = \41.60 .

TSIA2-M12-010 — Correct: D — Correct Answer: 2 — Explanation: The y-intercept gives the point (0, -6). Using (0, -6) and (4, 2), the slope is $(2 - (-6))/(4 - 0) = 8/4 = 2$.

TSIA2-M12-011 — Correct: B — Correct Answer: 18π centimeters — Explanation: The circumference of a circle is $C = 2\pi r$. With $r = 9$, $C = 18\pi$ centimeters.

TSIA2-M12-012 — Correct: C — Correct Answer: 51 — Explanation: The range is the greatest value minus the least value: $183 - 132 = 51$.

TSIA2-M12-013 — Correct: C — Correct Answer: 170 — Explanation: Compute 68% of 250: $0.68 \times 250 = 170$.

TSIA2-M12-014 — Correct: B — Correct Answer: (3, 5) — Explanation: Substitute (3, 5): $2(3) - 5 = 1$ and $3 + 5 = 8$, so the ordered pair satisfies both equations.

TSIA2-M12-015 — Correct: C — Correct Answer: 18 feet — Explanation: The ramp is the hypotenuse of a right triangle. The height is $\sqrt{(30^2 - 24^2)} = \sqrt{(900 - 576)} = \sqrt{324} = 18$ feet.

TSIA2-M12-016 — Correct: B — Correct Answer: 15 — Explanation: A mean of 14 for 9 numbers gives a total of $9 \times 14 = 126$. Adding 24 gives 150 for 10 numbers, and $150 \div 10 = 15$.

TSIA2-M12-017 — Correct: A — Correct Answer: \$21.75 — Explanation: A 6-pound package has 1 first pound and 5 additional pounds. The total charge is $\$9.75 + 5(\$2.40) = \$9.75 + \$12.00 = \$21.75$.

TSIA2-M12-018 — Correct: B — Correct Answer: $(n - 5)(n + 3)$ — Explanation: “5 less than a number n ” is $n - 5$, and “3 more than the same number” is $n + 3$. Their product is $(n - 5)(n + 3)$.

TSIA2-M12-019 — Correct: B — Correct Answer: 68° — Explanation: The angles in a triangle sum to 180° . The third angle is $180^\circ - 38^\circ - 74^\circ = 68^\circ$.

TSIA2-M12-020 — Correct: B — Correct Answer: Exactly 36% of customers chose water. — Explanation: Water was chosen by 72 out of 200 customers, and $72/200 = 0.36$, or 36%.

